

Ex.no:8	MINI PROJECT – SOCIAL MEDIA DATA ANALYTICS

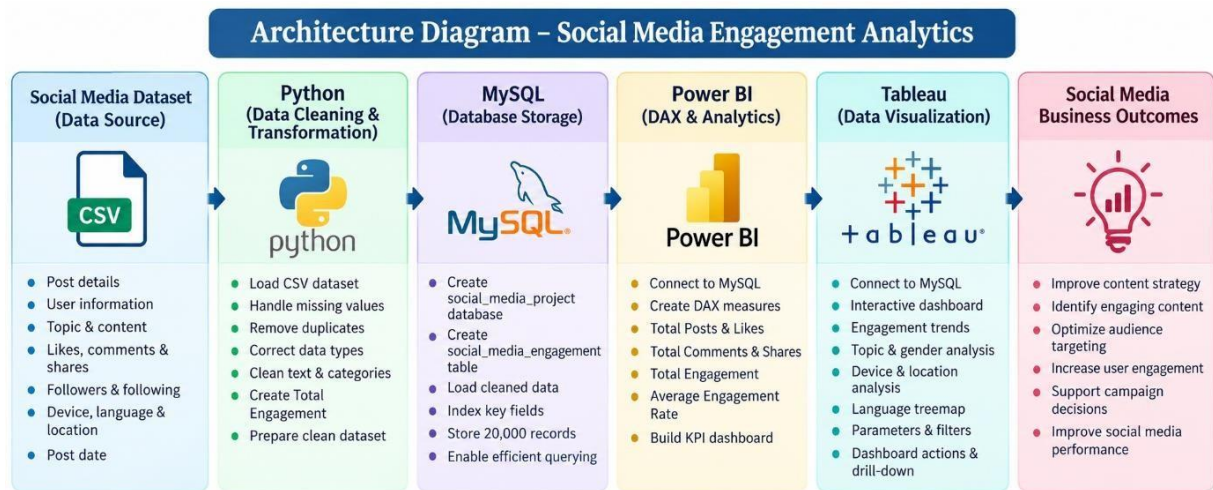
Aim:

To clean, transform, store, and analyze Social Media Engagement data using Python, MySQL, Power BI, and Tableau to derive meaningful insights into post performance, user engagement, user demographics, content topics, devices, locations, languages, engagement trends, and key engagement metrics.

Algorithm:

1. **Load** the Social Media Engagement dataset from the CSV file using **Pandas**.
2. **Clean** the data by handling missing values, removing duplicate records, correcting data types, and eliminating unnecessary attributes.
3. **Transform** the data by standardizing column names and preparing useful fields such as Total Engagement, Engagement Rate, Engagement per Follower, Post Date, User Age, Followers Count, Likes, Comments, and Shares.
4. **Store** the cleaned and transformed dataset in a **MySQL database** named `social_media_project`.
5. **Connect** the MySQL database to **Power BI** and **Tableau** for data analysis and visualization.
6. **Create DAX measures** in Power BI for key metrics such as Total Posts, Total Likes, Total Comments, Total Shares, Total Engagement, and Average Engagement Rate.
7. Develop interactive visualizations to analyze engagement trends, topic-wise engagement, gender-wise engagement, device-wise engagement, location-wise engagement, language-wise engagement, likes, comments, shares, and overall post performance.
8. Implement advanced Tableau features such as calculated fields, parameters, filters, interactive dashboards, dashboard actions, drill-down, and advanced visualizations including line charts, bar charts, pie/donut charts, maps, and treemaps.
9. **Analyze and interpret** the visualizations to identify high-performing topics, user engagement patterns, popular devices, geographic trends, and changes in engagement across different metrics.
10. Generate meaningful business insights from the Power BI and Tableau dashboards to support data-driven understanding of social media performance.
11. **End** the process.

Architecture:



Program:

```
import pandas as pd
```

```
# Load dataset
```

```
df = pd.read_csv("synthetic_social_media_engagement.csv")
```

```
# Create a copy for processing
```

```
social_media = df.copy()
```

```
# Remove duplicate records
```

```
social_media.drop_duplicates(inplace=True)
```

```
# Handle missing values
```

```
social_media["user_name"] = social_media["user_name"].fillna("Unknown")
```

```
social_media["location"] = social_media["location"].fillna("Unknown")
```

```
social_media["topic"] = social_media["topic"].fillna("Unknown")
```

```
social_media["device"] = social_media["device"].fillna("Unknown")
```

```
social_media["language"] = social_media["language"].fillna("Unknown")
```

```
social_media["hashtags"] = social_media["hashtags"].fillna("No Hashtags")
```

```
# Convert date columns
```

```
social_media["account_creation_date"] = pd.to_datetime(  
    social_media["account_creation_date"],  
    errors="coerce"  
)
```

```
social_media["post_date"] = pd.to_datetime(  
    social_media["post_date"],  
    errors="coerce"  
)
```

```
# Convert numerical columns
```

```
numeric_columns = [  
    "user_age",  
    "followers_count",  
    "following_count",  
    "content_length",  
    "likes",  
    "comments",  
    "shares",  
    "engagement_rate"  
]
```

```
for column in numeric_columns:
```

```
    social_media[column] = pd.to_numeric(  
        social_media[column],  
        errors="coerce"  
    )
```

```
# Fill missing numerical values
```

```
for column in numeric_columns:
    social_media[column] = social_media[column].fillna(0)
```

```
# Create derived metric
```

```
social_media["total_engagement"] = (
    social_media["likes"]
    + social_media["comments"]
    + social_media["shares"]
)
```

```
# Create engagement per follower
```

```
social_media["engagement_per_follower"] = (
    social_media["total_engagement"]
    / social_media["followers_count"].replace(0, 1)
)
```

```
# Standardize column names
```

```
social_media.columns = (
    social_media.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)
```

```
# Display dataset information
```

```
print("Original Shape:", df.shape)
print("Cleaned Shape:", social_media.shape)
```

```
print("\nCleaned Dataset:")
```

```
print(social_media.head())
```

```
print("\nMissing Values:")
```

```
print(social_media.isnull().sum())
```

```
print("\nDuplicate Records:")
```

```
print(social_media.duplicated().sum())
```

2. MySQL – Database and Data Loading

MySQL Command Prompt

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL

```
CREATE DATABASE IF NOT EXISTS social_media_project;
```

```
USE social_media_project;
```

Python / Jupyter

```
from sqlalchemy import create_engine
```

```
import pandas as pd
```

```
# MySQL connection details
```

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(
```

```
    f"mysql+pymysql://{username}:{password}@{host}:{port}/social_media_project"
```

```
)
```

```
# Load cleaned data into MySQL
```

```
social_media.to_sql(
```

```
    "social_media_engagement",
```

```
    con=db_engine,
```

```
    if_exists="replace",
```

```

index=False,

chunksize=5000

)

print("Social Media Engagement data loaded successfully into MySQL.")

Verify Data

pd.read_sql(

    "SELECT COUNT(*) AS total_rows FROM social_media_engagement",

    db_engine

)

```

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

1  -- Create database
2  CREATE DATABASE IF NOT EXISTS social_media_project;
3  USE social_media_project;
4
5  -- View tables
6  SHOW TABLES;
7
8  -- Check total records
9  SELECT COUNT(*) AS total_records
10 FROM social_media_engagement;
11
12 -- View sample data
13 SELECT * FROM social_media_engagement LIMIT 1000;

```

The Results grid shows the output of the query, displaying 1000 rows of sample data. The columns include post_id, user_id, user_name, user_gender, user_age, followers_count, following_count, account_creation_date, is_verified, location, topic, post_content, content_length, hashtags, has_media, post_date, post_id, device, and language.

The Output tab shows the execution log with the following actions:

#	Time	Action	Message	Duration / Fetch
1	17:12:03	CREATE DATABASE IF NOT EXISTS social_media_project	1 row(s) affected	0.016 sec
2	17:12:04	USE social_media_project	0 row(s) affected	0.000 sec
3	17:12:04	SHOW TABLES	1 row(s) returned	0.000 sec
4	17:12:05	SELECT COUNT(*) AS total_records FROM social_media_engagement	1 row(s) returned	0.000 sec
5	17:12:05	SELECT * FROM social_media_engagement LIMIT 1000	1000 row(s) returned	0.031 sec

3. Power BI – DAX Measures

1. Total Posts

Total Posts = `DISTINCTCOUNT(social_media_engagement[post_id])`

2. Total Likes

Total Likes = `SUM(social_media_engagement[likes])`

3. Total Comments

Total Comments = `SUM(social_media_engagement[comments])`

4. Total Shares

Total Shares = `SUM(social_media_engagement[shares])`

5. Total Engagement

Total Engagement =

[Total Likes] + [Total Comments] + [Total Shares]

6. Average Engagement Rate

Average Engagement Rate =

AVERAGE(social_media_engagement[engagement_rate])

7. Average Followers

Average Followers =

AVERAGE(social_media_engagement[followers_count])

8. Average Content Length

Average Content Length =

AVERAGE(social_media_engagement[content_length])

9. Verified Users

Verified Users =

```
CALCULATE(  
    DISTINCTCOUNT(social_media_engagement[user_id]),  
    social_media_engagement[is_verified] = "True"  
)
```

10. Posts with Media

Posts with Media =

```
CALCULATE(  
    DISTINCTCOUNT(social_media_engagement[post_id]),  
    social_media_engagement[has_media] = "True"  
)
```

4. POWER BI Dashboard

Created visuals in Power Bi in separate sheets and created a dashboard

Original rows: 20000
Cleaned rows: 20000
Columns: 22

Missing values:
post_id 0
user_id 0
user_name 0
user_gender 0
user_age 0
followers_count 0
following_count 0
account_creation_date 0
is_verified 0
location 0
topic 0
post_content 0
content_length 0
hashtags 0
has_media 0
post_date 0
device 0
language 0
likes 0
comments 0
shares 0
engagement_rate 0
dtype: int64

	post_id	user_id	user_name	user_gender	user_age	followers_count	following_count	account_creation_date	is_verified	location	topic	post_content	content_length	hashtags	has_media	post_date	device	language	likes	comments	shares	engagement_rate
0	P001A7K9	U1001	@travelwithme	Female	24	12500	320	2021-06-14	True	Mumbai	Travel	Exploring the beaches of Goa!	42	#travel #goa	True	2024-05-12	Mobile	English	1250	230	120	0.128
1	P002B3L4	U1002	@techworld	Male	28	50200	410	2019-11-03	False	Bengaluru	Technology	New smartphone launch looks amazing!	37	#tech #smartphone	True	2024-05-13	Desktop	English	3420	560	310	0.085
2	P003C8M1	U1003	@fashionista	Female	22	18700	980	2022-01-25	True	Delhi	Fashion	Summer collection is finally here	35	#fashion #style	False	2024-05-13	Mobile	Hindi	980	143	90	0.064
3	P004D6Q2	U1004	@sportsfan	Male	27	30800	760	2020-09-17	False	Chennai	Sports	What a match! India wins again!	33	#cricket #india	True	2024-05-14	Mobile	English	4120	620	450	0.112
4	P005E9Z8	U1005	@foodieats	Female	25	22100	540	2021-03-10	True	Hyderabad	Food	Tried this amazing biryani today!	37	#food #biryani	True	2024-05-14	Mobile	Tamil	2890	410	230	0.096

In [10]:

```
pd.read_sql(  
    "SELECT COUNT(*) AS total_rows FROM social_media_engagement",  
    db_engine  
)
```

Out[10]:

	total_rows
0	20000

In [11]:

```
pd.read_sql(  
    "SELECT * FROM social_media_engagement LIMIT 5",  
    db_engine  
)
```

Out[11]:

	post_id	user_id	user_name	user_gender	user_age	followers_count	following_count	account_creation_date	is_verified	location	topic	post_content	content_length	hashtags	has_media	post_date	device	language	likes	comments	shares	engagement_rate
0	P001A7K9	U1001	@travelwithme	Female	24	12500	320	2021-06-14	True	Mumbai	Travel	Exploring the bea...	42	#travel #goa	True	2024-05-12	Mobile	English	1250	230	120	0.128
1	P002B3L4	U1002	@techworld	Male	28	50200	410	2019-11-03	False	Bengaluru	Technology	New smartphone L...	37	#tech #smartpho...	True	2024-05-13	Desktop	English	3420	560	310	0.085
2	P003C8M1	U1003	@fashionista	Female	22	18700	980	2022-01-25	True	Delhi	Fashion	Summer collection...	35	#fashion #style	False	2024-05-13	Mobile	Hindi	980	143	90	0.064
3	P004D6Q2	U1004	@sportsfan	Male	27	30800	760	2020-09-17	False	Chennai	Sports	What a match! In...	33	#cricket #india	True	2024-05-14	Mobile	English	4120	620	450	0.112
4	P005E9Z8	U1005	@foodieats	Female	25	22100	540	2021-03-10	True	Hyderabad	Food	Tried this amazin...	37	#food #biryani	True	2024-05-14	Mobile	Tamil	2890	410	230	0.096

5 rows x 22 columns



Navigator

Display Options ▾

localhost:3306; social_media_project [1]

social_media_engagement

social_media_engagement

post_id	user_id	user_name	user_gender	user_age	followers_count	following_count	account_creation_date	is_verified	location
P001A7K9	U1001	@travelwithme	Female	24	12500	320	2021-06-14	True	Mumbai
P002B3L4	U1002	@techworld	Male	28	50200	410	2019-11-03	False	Bengaluru
P003C8M1	U1003	@fashionista	Female	22	18700	980	2022-01-25	True	Delhi
P004D6Q2	U1004	@sportsfan	Male	27	30800	760	2020-09-17	False	Chennai
P005E9Z8	U1005	@foodieeats	Female	25	22100	540	2021-03-10	True	Hyderabad
P006F4R1	U1006	@musiclover	Male	30	63400	890	2018-07-21	True	Kolkata
P007G5T3	U1007	@moviebuff	Female	26	29800	660	2020-12-05	False	Pune
P008H7V2	U1008	@lifestylehub	Male	29	41200	520	2019-08-19	True	Chennai
P009J1X4	U1009	@gamerzone	Male	21	17500	310	2022-05-11	False	Mumbai
P010K3Y6	U1010	@bookworm	Female	23	26000	470	2021-09-28	True	Bengaluru

The preview has been truncated due to size limits.

Select Related Tables

Load

Transform Data

Cancel

File	Home	Help	Table tools	Share
Name	SocialMediaData	Manage relationships	New measure	Quick measure
New column	New table	Mark as date table		
Auto recovery contains some recovered files that haven't been opened.				
View recovered files				
Data				
Search				
SocialMediaData				
account_creation_date				
Average Engagement Rate				
comments				
content_length				
device				
engagement_rate				
followers_count				
following_count				
has_media				
hashtags				
is_verified				
language				
likes				
location				
post_content				
post_date				
post_id				
shares				
topic				
Total Comments				
Total Followers				
Table: SocialMediaData (20,000 rows)				

Data Visualization Using Power BI

The Social Media Engagement dataset was imported into Power BI Desktop to analyze post performance, user engagement, demographics, devices, locations, languages, and engagement trends.

Interactive KPI cards and charts were created to identify popular topics, user preferences, and overall social media performance.

Visualization 1 — Engagement by Topic Chart: Bar Chart

Shows the total engagement for different social media topics.

This visualization helps identify the most engaging content categories, such as Technology, Travel, Sports, Food, and Fashion.

Visualization 2 — Engagement by Location Chart: Map

Shows the distribution of social media engagement across different cities in India.

This visualization helps identify locations with the highest user engagement.

Visualization 3 — Gender-wise Engagement Chart: Bar Chart

Shows the comparison of engagement between different user genders:

- Male
- Female

This helps analyze whether engagement patterns differ based on gender.

Visualization 4 — Device-wise Engagement Chart: Pie Chart

Shows the distribution of social media engagement based on the devices used:

- Mobile
- Desktop
- Tablet

This visualization helps identify the most commonly used devices for social media activity.

Visualization 5 — Language-wise Engagement Chart: Treemap

Shows engagement across different languages used in social media posts.

This helps identify popular languages and understand audience preferences.

Visualization 6 — Engagement Trend Chart: Line Chart

Shows how social media engagement changes over time.

This visualization helps identify increasing or decreasing engagement patterns across different dates or months.

Visualization 7 — Total Engagement KPI Card: Card

Displays the total engagement generated through likes, comments, and shares.

This provides an overall measure of social media performance.

Visualization 8 — Average Engagement Rate KPI Card: Card

Shows the average engagement rate of all posts.
This helps evaluate the overall effectiveness of user interaction with the content.

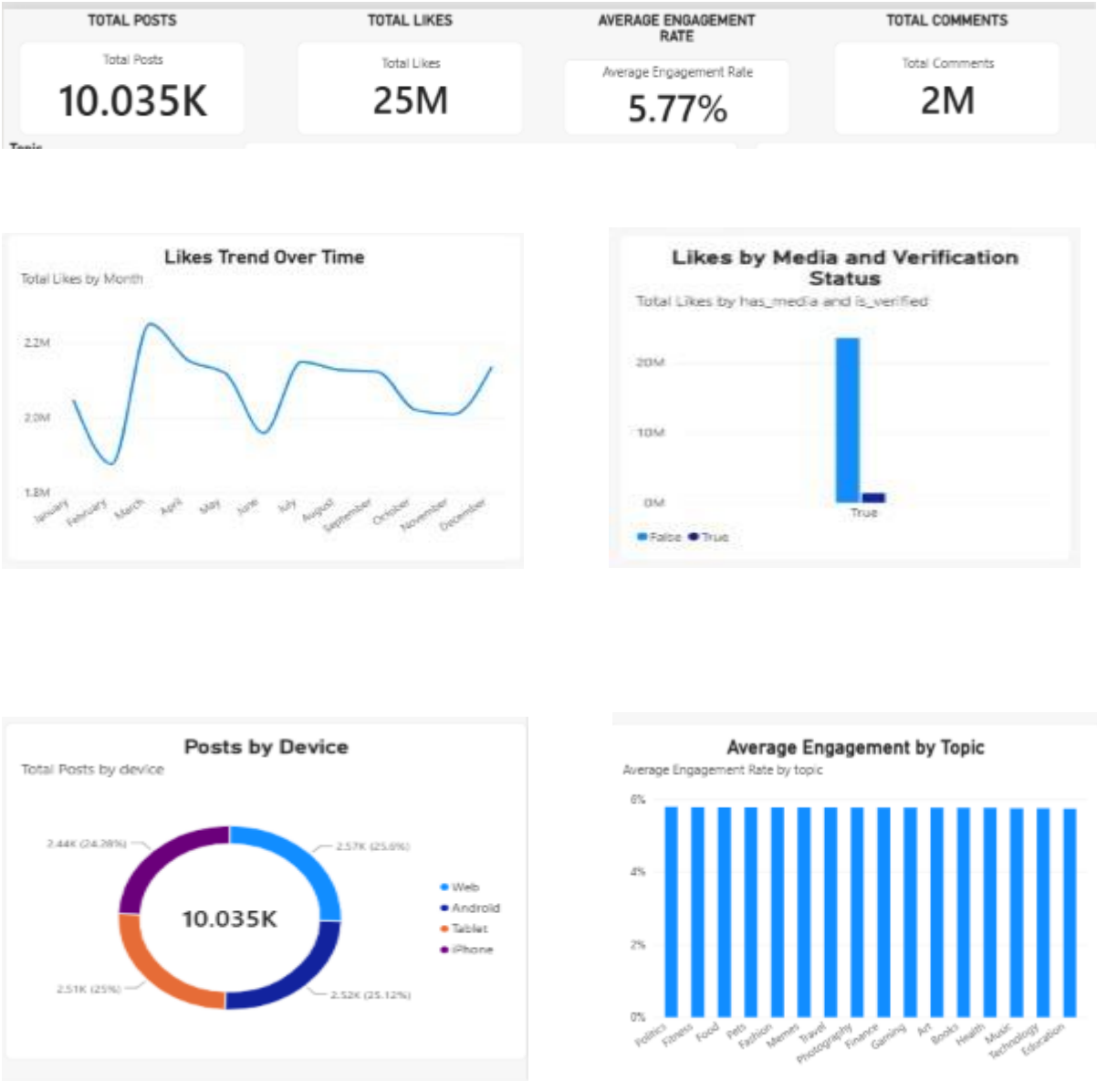
Visualization 9 — Total Posts KPI Card: Card

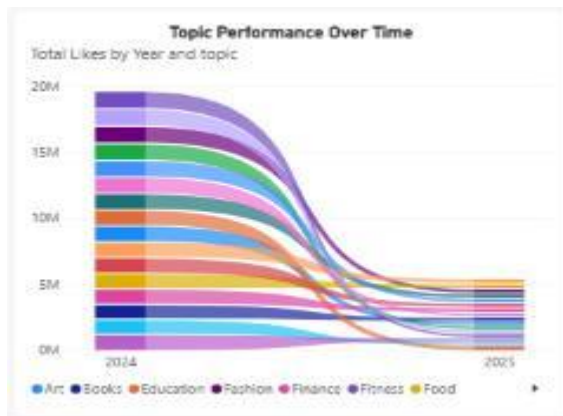
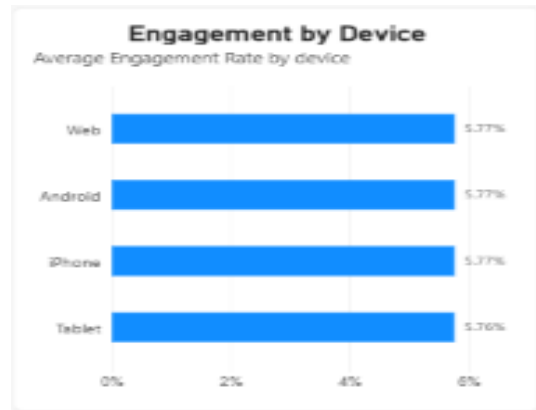
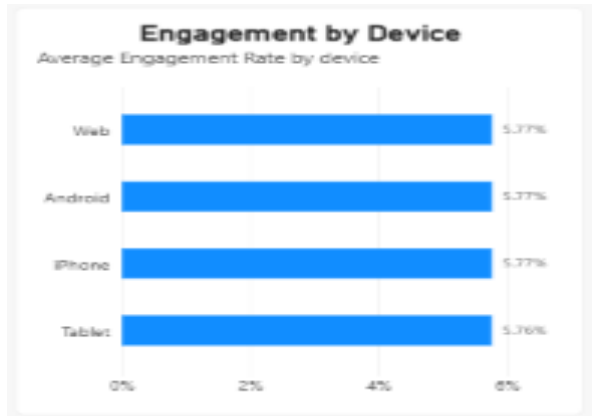
Displays the total number of social media posts in the dataset.
This provides an overview of the dataset size and posting activity.

Visualization 10 — Total Likes, Comments and Shares KPI Cards: Cards

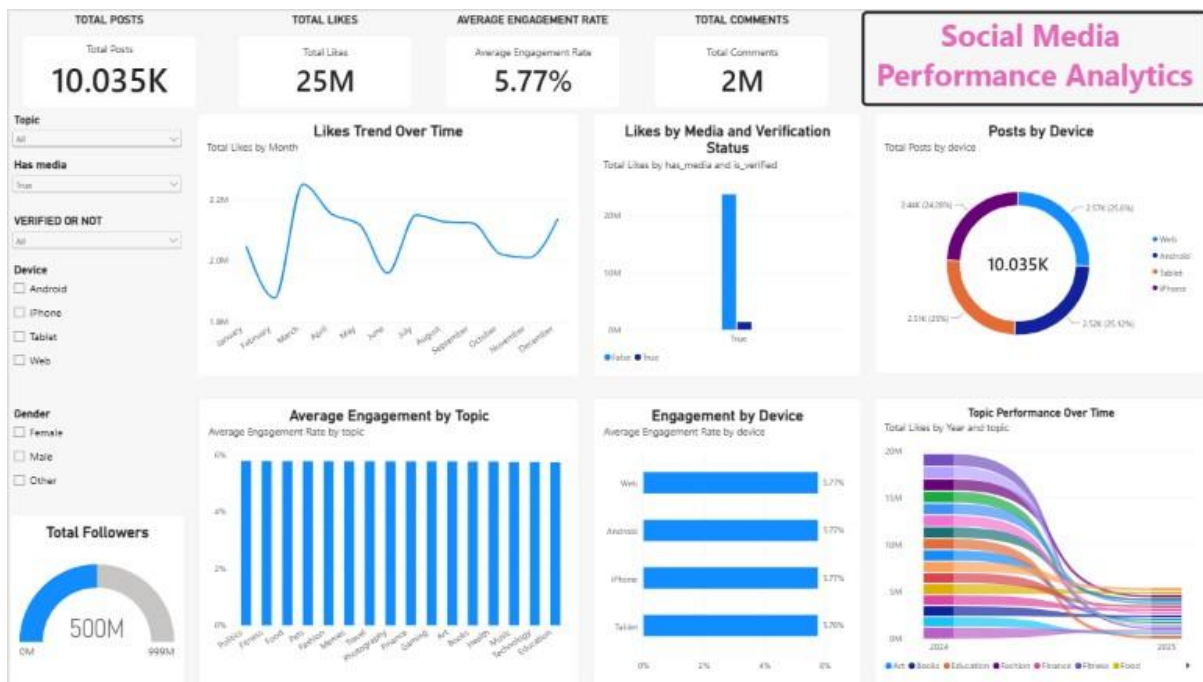
Shows the total number of likes, comments, and shares received by all posts.

OUTPUT :





SOCIAL MEDIA ANALYTICS POWER BI DASHBOARD

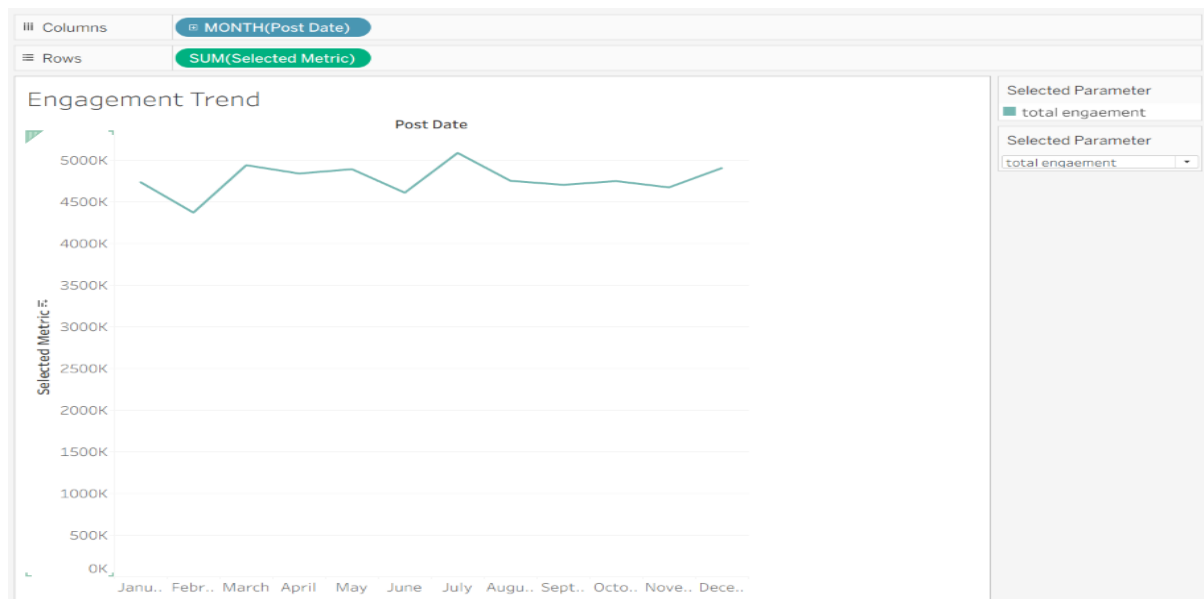


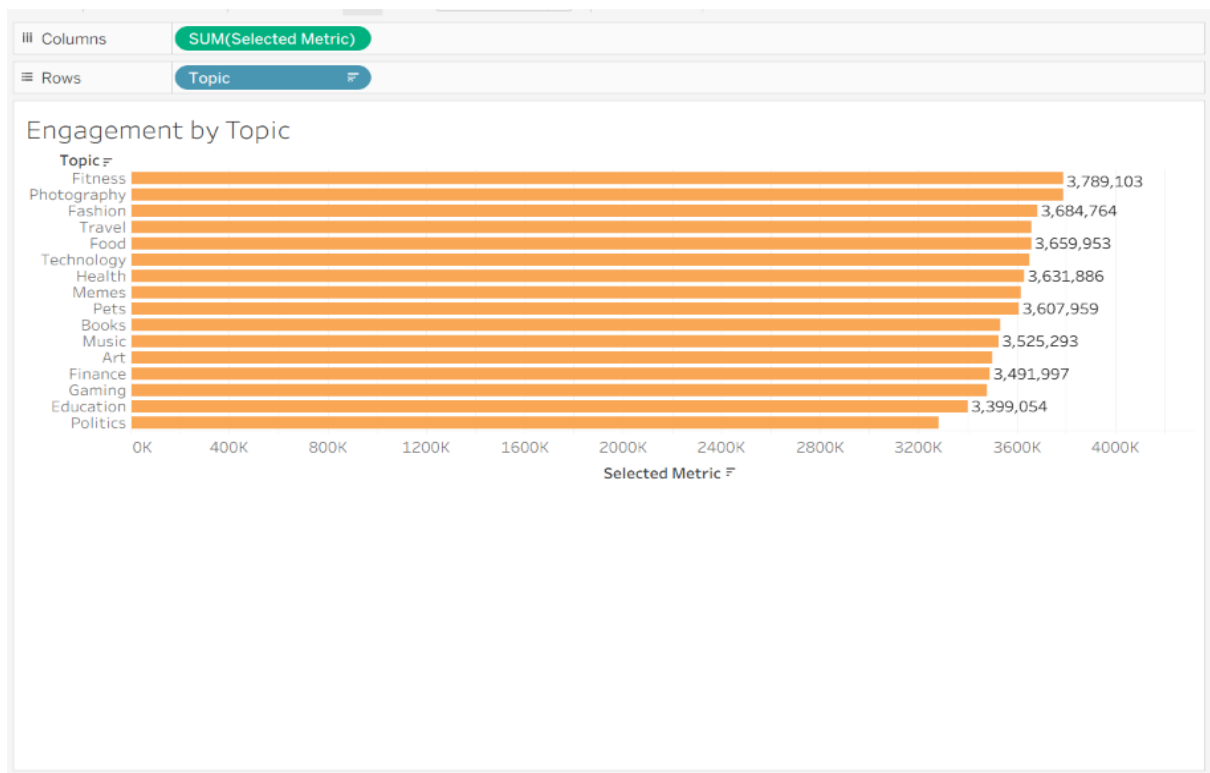
5. Tableau Dashboard

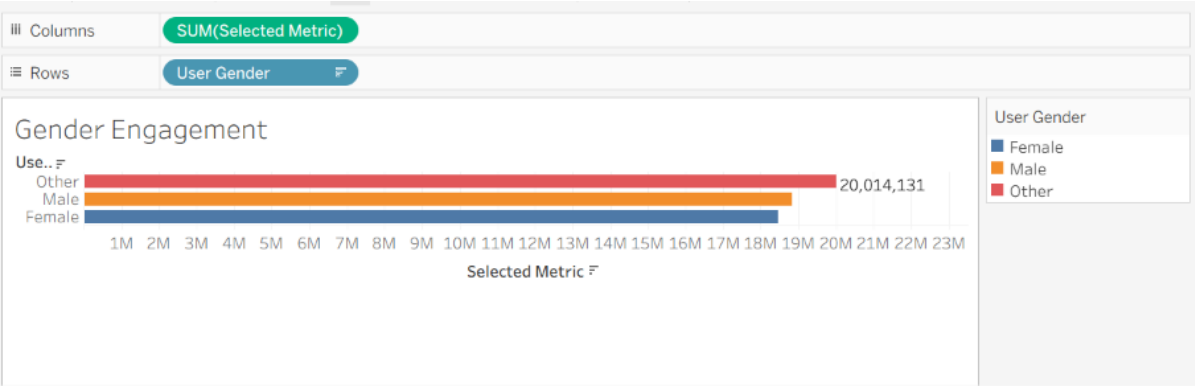
Created visuals in Tableau in separate sheets and created a dashboard

The screenshot shows the Tableau interface with the 'social_media_engagement' table selected. The sidebar on the left shows the 'Connections' list with 'social_media_engagement' (MySQL) selected, the 'Database' dropdown set to 'social_media_project', and the 'Table' dropdown set to 'social_media_engagement'. The main view displays the table's fields and a preview of 10 rows of data.

post_id	user_id	user_name	user_gender	user_age	followers_count	account_creation_date	is_verified	location	topic
P001A7K9	U1001	@travelwithme	Female	24	12500	2021-06-14	True	Mumbai	Travel
P002B3L4	U1002	@techworld	Male	28	50200	2019-11-03	False	Bengaluru	Technology
P003C8M1	U1003	@fashionista	Female	22	18700	2022-01-25	True	Delhi	Fashion
P004D6Q2	U1004	@sportsfan	Male	27	30800	2020-09-17	False	Chennai	Sports
P005E9Z8	U1005	@foodieeats	Female	25	22100	2021-03-10	True	Hyderabad	Food
P006F4R1	U1006	@musiclover	Male	30	63400	2018-07-21	True	Kolkata	Music
P007G5T3	U1007	@moviebuff	Female	26	29800	2020-12-05	False	Pune	Movies
P008H7V2	U1008	@lifestylehub	Male	29	41200	2019-08-19	True	Chennai	Lifestyle
P009J1K4	U1009	@gamerzone	Male	21	17500	2022-05-11	False	Mumbai	Gaming
P010K3Y6	U1010	@bookworm	Female	23	26000	2021-09-28	True	Bengaluru	Education

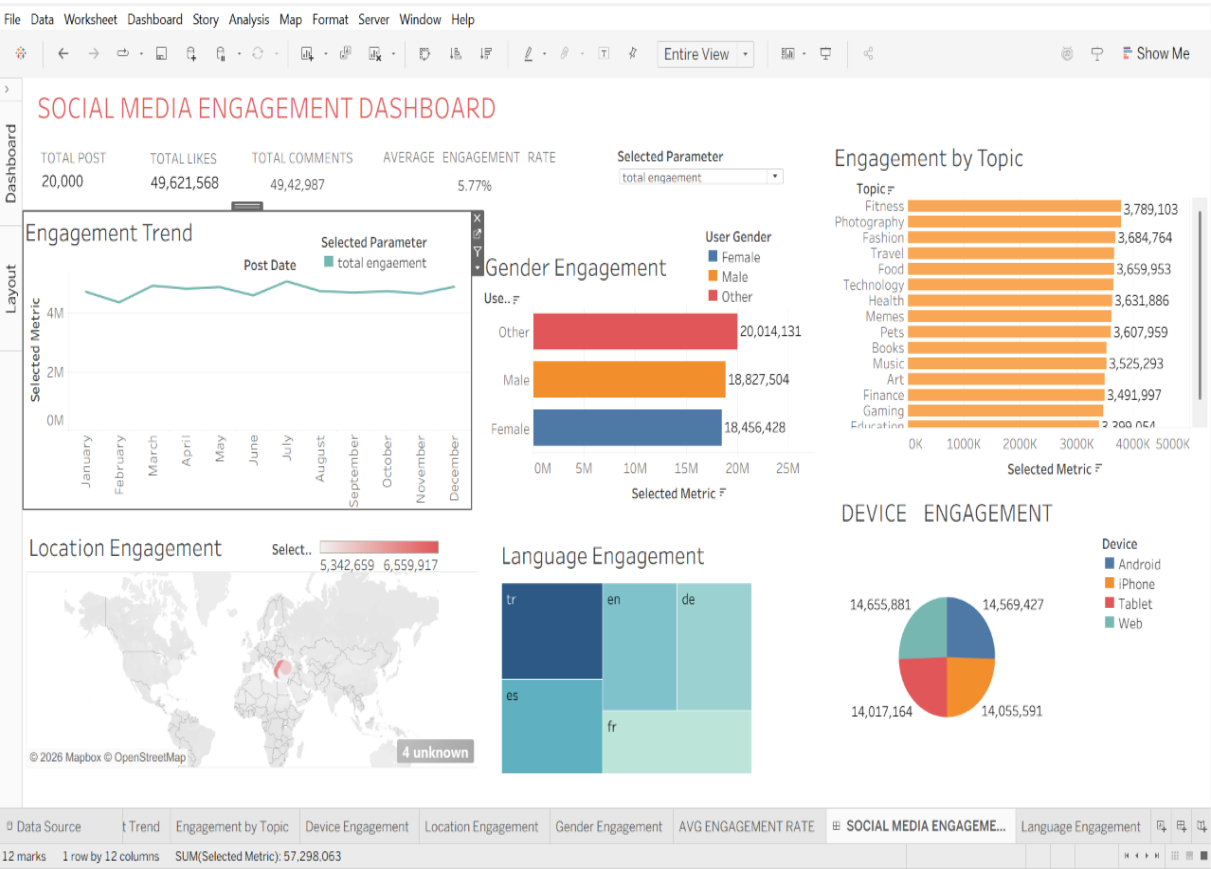






OUTPUT :

SOCIAL MEDIA ANALYTICS TABLEAU DASHBOARD



Result: